Curriculum Vitae

Hyeonsu Lee, Ph.D.

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EDUCATION

2018.02 ~ 2024.08	**Ph.D.**	Bio and Brain Engineering (Advisor: **Se-Bum Paik**)	**KAIST**	Korea
2016.03 ~ 2018.02	**M.S.**	Bio and Brain Engineering (Advisor: **Se-Bum Paik**)	**KAIST**	Korea
2011.02 ~ 2016.02	**B.S.**	Mathematical Sciences Business and Technology Management (Minor)	**KAIST**	Korea

RESEARCH EXPERIENCE

2024.10 ~ Present	Postdoctoral researcher (Advisor: **Jaeson Jang**)	**KIST**	- **Multi-agent model simulation** (LLM, agent-based, etc.) - **Data analysis** (human/animal behaviors, mouse proteomics)
2016.03 ~ 2024.08	Graduate researcher	**KAIST**	- **Neural circuit modeling** of visual cognitive functions - Interpret behavioral data via **reinforcement learning** - **Neural data analysis**: Spike, LFP, Calcium imaging, etc.
2015.01 ~ 2016.02	Research intern	**KAIST**	- Modeling visual system-like image compression algorithm
2014.04 ~ 2014.11	Researcher	**ybrain**	- Design clinical trials (AD and MDD), Neural data analysis, Human psychophysical experiments
2013.07 ~ 2013.08	Research intern	**KAIST**	- Analyzing functional brain imaging (DWI)

PUBLICATIONS

Neural Network Modeling

Lee H*, Choi W*, Lee D & Paik SB — ***Cell Reports*** (2023) [link]
"Comparison of visual quantities in untrained neural networks"

Lee H, Choi W, Park Y & Paik SB — ***Neural Networks*** (2020) [link]
"Distinct role of flexible and stable encodings in sequential working memory"

Cho S, **Lee H**, Baek S & Paik SB — ***Neural Networks*** (2025) [link]
"Neuromimetic metaplasticity for adaptive continual learning without catastrophic forgetting"

Neural Data Analysis

Jeong M*, **Lee H***, …, Paik SB†, Lim BK† & Kim D† — ***eNeuro*** (2021) [link]
"Interhemispheric cortico-cortical pathway for sequential bimanual movements in mice"

Bae JW*, Jeong H*, …, **Lee H**, Paik SB & Jung MW — ***Nature Communications*** (2021) [link]
"Parallel processing of working memory and temporal information by distinct types of cortical projection neurons"

Song M*, Kang MS*, **Lee H**, Jeong Y & Paik SB — ***Scientific Reports*** (2018) [link]
"Classification of spatiotemporal neural activity patterns in brain imaging data"

PRE-PRINTS

Choi W, **Lee H** & Paik SB — ***bioRxiv*** (2020) [link]
"Slow rhythmic eye motion predicts periodic alternation of bistable perception"

(*: equal contribution)

SELECTED CONFERENCE PRESENTATIONS

1 *Society for Neuroscience* (**SfN**) 2025, **Lee H**, Baek S, Choi W, Jang J & Paik SB,
"Spontaneous social tipping and trend cycles driven by randomness"

2 *Computational and Systems Neuroscience* (**Cosyne**) 2022, Choi W*, **Lee H*** & Paik SB,
"Spontaneous emergence of magnitude comparison units in untrained deep neural networks"

3 *Society for Neuroscience* (**SfN**) 2018, **Lee H**, Choi W, Park Y & Paik SB,
"Resource overwrite model for sequential working memory in human" (Oral)

4 *Korean Society for Brain and Neural Sciences* (**KSBNS**) 2017, **Lee H**, Choi W & Paik SB,
"Working memory pre-allocation improves the performance of sequential memory task" (Best Presentation Award)

PATENTS

Paik SB, **Lee H**, Park Y, **US Patent**, US 11,467,728 B2 (2022) "Storage device using neural network and operating method for automatic redistribution of information and variable storage capacity based on accuracy-storage capacity tradeoff thereof"
* 신경망을 이용한 기억 장치 및 그의 정확성-저장용량 트레이드오프에 기반한 정보의 자동적 재분배 및 가변적 용량 구현을 위한 동작 방법

Paik SB, Cho S, **Lee H**, Baek S, KR Patent, 10-2024-0199987 (2024) "Method for addressing catastrophic forgetting in artificial neural networks using neuromimetic metaplasticity rules observed in natural brains"
* 생물학적 두뇌의 메타가소성 법칙을 적용하여 인공신경망 학습에서 파괴적 망각을 해결하는 방법 및 장치

AWARDS

Poster Presentation Awards, Brain Science Institute of KIST, 2025

Best Presentation Awards, Korean Society for Brain and Neural Sciences (KSBNS), 2017

Poster Presentation Awards, Korean Society for Computational Neuroscience (cbrain), 2017

Best TA awards, Bio and Brain Engineering, KAIST, 2018

SKILLS

Brain-inspired AI, Python, MATLAB, Illustrator,

Neural network modeling & simulation, Data analysis, Mathematical modeling,

LLM simulation, Multi-agent simulation, Human psychophysical experiment, Eye-tracking

CONFERENCE PRESENTATIONS

International

1 *Korean Society for Brain and Neural Sciences* (**KSBNS**) 2025, **Lee H**, Baek S, Jang J & Paik SB,
"Spontaneous social change driven by Stochastic Individual Behavior"

2 *Society for Neuroscience* (**SfN**) 2023, **Lee H** & Paik SB,
"Emergence of aesthetic preferences in untrained deep neural networks"

3 *Society for Neuroscience* (**SfN**) 2023, Kim G, Kang M, **Lee H**, Park Y, Song M & Paik SB,
"Random imitation as an efficient survival strategy in multi-agent systems"

4 *Korean Society for Brain and Neural Sciences* (**KSBNS**) 2023, Kang M, Kim G, **Lee H**, Park Y, Song M & Paik SB, "Random imitation as an evolutionary strategy in a multi-agent system"

5 *Society for Neuroscience* (**SfN**) 2022, **Lee H***, Choi W*, Lee D & Paik SB, "Quantity comparison in untrained deep neural networks"

6 *Society for Neuroscience* (**SfN**) 2022, Kang M, **Lee H**, Park Y, Song M & Paik SB, "Random imitation for evolution in multi-agent systems"

7 *Annual Computational Neuroscience Meeting* (**CNS**) 2022, **Lee H***, Choi W*, Lee D & Paik SB, "Comparison of visual quantities in untrained deep neural networks"

8 *Korean Society for Brain and Neural Sciences* (**KSBNS**) 2022, **Lee H***, Choi W*, Lee D & Paik SB, "Visual quantity comparison in untrained deep neural networks"

9 *Society for Neuroscience* (**SfN**) 2021, **Lee H***, Choi W* & Paik SB, "Visual proportion sense in untrained deep neural networks"

10 *Korean Society for Brain and Neural Sciences* (**KSBNS**) 2020, Bae JW, …, **Lee H,** Paik SB & Jung MW, "Distinct types of cortical projection neurons for working memory and timing"

11 *Annual Computational Neuroscience Meeting* (**CNS**) 2020, Choi W, **Lee H** & Paik SB, "Rhythmic eye movement predicts active perception of ambiguous visual stimulus"

12 *Society for Neuroscience* (**SfN**) 2019, Jeong M, **Lee H**, …, Lim BK, Paik SB & Kim D, "Neuronal ensemble encoding bimanual motor coordination in mice"

13 *Annual Computational Neuroscience Meeting* (**CNS**) 2018, **Lee H**, Choi W, Park Y & Paik SB, "A dynamic resource model for sequential working memory"

14 *Society for Neuroscience* (**SfN**) 2017, **Lee H***, Park Y* & Paik SB, "Pre-allocation of memory resources improves working memory performance in a sequential memory task"

15 *Annual Computational Neuroscience Meeting* (**CNS**) 2017, **Lee H**, Choi W & Paik SB, "Pre-allocation of working memory modulates memory performance"

16 *Society for Neuroscience* (**SfN**) 2016, Choi W, **Lee H** & Paik SB, "Local disinhibition modulates brain functions by mediating global network state"

17 *Society for Neuroscience* (**SfN**) 2016, Song M*, Kang M*, **Lee H**, Lee H, Jeong Y† & Paik SB†, "Spatio-Temporal Classification of Neural Activity Patterns in Dynamic Brain Imaging Data"

18 *Annual Computational Neuroscience Meeting* (**CNS**) 2016, Song M, **Lee H** & Paik SB, "Quantitative Classification of Neural Network Activity Patterns in Imaging Data"

Domestic

1 *Korean Society for Computational Neuroscience Meeting* (**cbrain**) 2025, **Lee H**, Baek S, Jang J & Paik SB, "Emergent Cycles in Social Behavior from Stochastic Individuals"

2 *Korean Society for Cognitive Science Meeting* (**KSCS**) 2018, **Lee H**, Choi W, Park Y & Paik SB, "Sequential overwrite model can explain the serial-position effects in working memory"

3 *Korean Physical Society Meeting* (**KPS**) 2018, **Lee H**, Choi W, Park Y & Paik SB, "A theory of resource allocation for sequential memory in human"

4 *Annual Korean Society for Computational Neuroscience Meeting* (**cbrain**) 2017, **Lee H**, Choi W & Paik SB, "Working memory resource allocation improves the sequential memory performance"

5 *Korean Physical Society Meeting* (**KPS**) 2017, **Lee H**, Choi W & Paik SB, "A comparative study on simultaneous versus sequential presentation of stimuli for working memory performance"

6 *Korean Physical Society Meeting* (**KPS**) 2016, Song M, **Lee H** & Paik SB,
"Computational Classification of Neural Network Activity Patterns for Imaging Data"

7 *Korean Society for Computational Neuroscience Meeting* (**cbrain**) 2015, **Lee H***, Song M* & Paik SB,
"Classification of the Neural Activity Patterns by Spatio-Temporal Correlation Index"

8 *Korean Physical Society Meeting* (**KPS**) 2015, **Lee H**, Jang J & Paik SB,
"A Biologically Inspired Adaptive Model for Efficient Image Compression"